Climate

SWAT+ Documentation

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Purpose

The `.cli` files are index files. Each one names a set of climate inputs that SWAT+ should load. They sit between `file.cio`, which names the index files themselves, and the actual time series files (`*.pcp`, `*.tmp`, `*.slr`, `*.hmd`, `*.wnd`, `*.pet`, weather-generator stations, atmospheric deposition).

The full set of index files is defined by `type input_cli` in `src/input_file_module.f90`:

Slot	Default name	Reader	Purpose
`weat_sta`	`weather-sta.cli`	`cli_staread.f90`	Weather stations: one row per station, columns are pointers to a `wgn` station and to per-variable gages.
`weat_wgn`	`weather-wgn.cli`	`cli_wgnread.f90`	Weather generator parameters: location plus 12 monthly statistics.
`pet_cli`	`pet.cli`	`cli_petmeas.f90`	List of measured PET station files.
`pcp_cli`	`pcp.cli`	`cli_pmeas.f90`	List of measured precipitation station files.
`tmp_cli`	`tmp.cli`	`cli_tmeas.f90`	List of measured temperature station files.
`slr_cli`	`slr.cli`	`cli_smeas.f90`	List of measured solar radiation station files.
`hmd_cli`	`hmd.cli`	`cli_hmeas.f90`	List of measured relative humidity station files.
`wnd_cli`	`wnd.cli`	`cli_wmeas.f90`	List of measured wind speed station files.
`atmo_cli`	`atmodep.cli`	`cli_read_atmodep.f90`	Atmospheric N deposition: control record plus one or more stations.

In `file.cio` the slot value is either a filename or the literal token `null`. Every reader skips its work when the slot is `null` or when the file does not exist. The per-variable readers (`pcp.cli`, `tmp.cli`, `slr.cli`, `hmd.cli`, `wnd.cli`, `pet.cli`) all have the same shape: title, header, then one filename per line. The actual time-series files are documented separately.

All readers skip a title line and a header line at the top of the file. Records are list-directed (free format), so columns are separated by whitespace.

weather-sta.cli

One record per weather station. Each row binds a station name to a weather generator and to gage names for each climate variable.

Source

- Reader: src/cli_staread.f90
- Type: src/climate_module.f90, `type weather_station` (column 1) and `type weather_codes_station_char` (columns 2 to 9)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Lines 3+: 9 fields per record, list-directed.

The reader call is `read (107,*,iostat=eof) wst(i)%name, wst(i)%wco_c`. The fields of `wco_c` are filled in declaration order.

#	Column	Type	Description
1	name	char(50)	Weather station name. Referenced by `hru.con` and other spatial-object files.
2	wgn	char(50)	Name of the weather generator station in `weather-wgn.cli`.
3	pgage	char(50)	Name of the precipitation gage in `pcp.cli`, or `sim` to generate.

#	Column	Type	Description
4	tgage	char(50)	Name of the temperature gage in `tmp.cli`, or `sim` to generate.
5	sgage	char(50)	Name of the solar radiation gage in `slr.cli`, or `sim` to generate.
6	hgage	char(50)	Name of the relative humidity gage in `hmd.cli`, or `sim` to generate.
7	wgage	char(50)	Name of the wind speed gage in `wnd.cli`, or `sim` to generate.
8	petgage	char(50)	Name of the measured PET station in `pet.cli`, or `null`.
9	atmodep	char(50)	Name of the atmospheric deposition station in `atmodep.cli`, or `null`.

Each gage name is matched against the station list of the corresponding `.cli` file. A missing match (and a value other than `sim` for the generated variables, or `null` for pet/atmodep) is written to the error log at unit 9001.

Example

`refdata/Ames_sub1/weather-sta.cli`:

`weather-sta.cli`:

name	wgn	pcp	tmp	slr	hm
wgn_sta1	wgn_001	ames.pcp	ames.tem	sim	si

The header column at position 8 is labelled `wnd_dir` by SWAT+ Editor. The reader stores that value in `petgage` (the PET gage slot).

weather-wgn.cli

One record per weather generator station. Each record is a station header line plus a 12-row monthly statistics block.

Source

- Reader: src/cli_wgnread.f90
- Type: src/climate_module.f90, `type weather_generator_db`

Format

- Line 1: title (skipped).
- Per station, 14 lines:
  - Line A: station name, latitude, longitude, elevation, rain_yrs.
  - Line B: header for the monthly block (skipped).
  - Lines C1..C12: one line per month, 14 values each.

Number of stations is determined by reading to end of file.

Station header line:

#	Field	Type	Units	Description
1	name	char	-	Generator station name. Matched by `wgn` in `weather-sta.cli`.
2	lat	real	degrees	Latitude of the generator station.
3	long	real	degrees	Longitude of the generator station.
4	elev	real	m	Elevation of the generator station.
5	rain_yrs	real	years	Number of years of record used to compute `rainhmx`.

Monthly block, 12 rows (Jan to Dec), 14 columns:

#	Field	Type	Units	Description
1	tmpmx	real	deg C	Average daily maximum air temperature.
2	tmpmn	real	deg C	Average daily minimum air temperature.
3	tmpstdmx	real	deg C	Standard deviation of daily maximum air temperature.
4	tmpstdmn	real	deg C	Standard deviation of daily minimum air temperature.
5	pcpmm	real	mm	Total precipitation in the month.
6	pcpstd	real	mm/day	Standard deviation of daily precipitation.
7	pcpskw	real	none	Skew coefficient of daily precipitation.
8	pr_wd	real	none	Probability of a wet day after a dry day.
9	pr_ww	real	none	Probability of a wet day after a wet day.
10	pcpd	real	days	Average number of days of precipitation in the month.
11	rainhmx	real	mm	Maximum 0.5-hour rainfall in the month.
12	solarav	real	MJ/m^2/day	Average daily solar radiation.
13	dewpt	real	deg C	Average daily dew point temperature.
14	windav	real	m/s	Average daily wind speed.

Example

`refdata/Ames_sub1/weather-wgn.cli`:

```
weather-wgn.cli: Ames
wgn001      37.48     -100.83   887.00    10
   TMPMX   TMPMN  TMPSTDMX TMPTDMN  PCPMM  PCPSTD  PCPSKW   PR_WD   PR_WW   PCPD  RAINHMX SOLARAV  DEWPT
    7.61   -8.13    8.06    5.99     7.4     3.3    0.19    0.07    0.25    2.65     5.1   10.39   -8.14
   11.07   -5.53    8.06    5.34    10.2     5.1    2.67    0.08    0.27    2.86     1.8   13.41   -5.58
   15.22   -1.87    7.99    5.22    28.1     8.9    1.64    0.11    0.34    4.43     7.6   17.85   -4.71
   21.42    4.12    6.42    4.82    32.0     7.9    1.65    0.14    0.34    5.25    24.6    21.7    0.93
   25.97    9.93    5.68    4.29    83.7    12.4    1.71    0.21    0.42    8.24    26.9   23.09    7.81
   31.78   15.51    4.96    3.62    71.6    13.0     0.9    0.17    0.41    6.71    40.4   26.15   12.49
   34.28   18.34    3.68    2.51    65.0    12.4    1.25    0.18    0.33    6.56    37.3   25.85   14.72
   33.17   17.22    3.89    2.59    65.0    13.0    1.26    0.19    0.27     6.4    32.0   22.54   14.29
   29.04   12.36    5.12    4.25    43.5    13.5    3.71    0.13    0.31    4.76    18.0   20.32    9.38
   23.43    5.54    6.17    4.63    29.8    11.9    1.31    0.08    0.32    3.26    10.4   15.04    3.48
   14.34   -1.71    6.76    4.75    20.1     6.9    0.58    0.09    0.34     3.6    10.4   15.46   -2.96
    9.17   -6.26    7.09    5.01     8.0     3.6    1.24    0.07    0.24    2.61    16.3    9.76   -6.38
```

This file holds one generator station (`wgn001`).

pcp.cli

List of measured precipitation station files. Each row points to one `*.pcp` file in the project directory (or under the path set by `in_path_pcp%pcp`).

Source

- Reader: src/cli_pmeas.f90
- Type: src/climate_module.f90, `type climate_measured_data` (`pcp` array)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Lines 3+: one field per record.

#	Column	Type	Description
1	filename	char(50)	Name of a `*.pcp` time-series file. The same string is used as the station key (`pcp_n(i)`) that `weather-sta.cli` matches against.

The reader makes two passes over the file. The first pass reads the field into `pcp_n(i)` (the lookup name). The second pass reads the same field into `pcp(i)%filename` and opens that file.

Example

`refdata/Ames_sub1/pcp.cli`:

```
pcp.cli
FILENAME
ames.pcp
```

One station, file `ames.pcp` in the project directory.

tmp.cli

List of measured temperature station files.

Source

- Reader: src/cli_tmeas.f90
- Type: src/climate_module.f90, `type climate_measured_data` (`tmp` array)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Lines 3+: one field per record.

#	Column	Type	Description
1	filename	char(50)	Name of a `*.tmp` time-series file. Used as the station key (`tmp_n(i)`).

Optional weather path prefix from `in_path_tmp%tmp` (see `file.cio`).

Example

`refdata/Ames_sub1/tmp.cli`:

```
tmp.cli
FILENAME
ames.tmp
```

slr.cli

List of measured solar radiation station files.

Source

- Reader: src/cli_smeas.f90
- Type: src/climate_module.f90, `type climate_measured_data` (`slr` array)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Lines 3+: one field per record.

#	Column	Type	Description
1	filename	char(50)	Name of a `*.slr` time-series file. Used as the station key (`slr_n(i)`).

Example

```
slr.cli
FILENAME
ames.slr
```

The reference dataset `refdata/Ames_sub1` does not ship a `slr.cli` because the weather-station rows use `sim` for solar radiation. The example shape above is what the reader expects when the file is present.

hmd.cli

List of measured relative humidity station files.

Source

- Reader: src/cli_hmeas.f90
- Type: src/climate_module.f90, `type climate_measured_data` (`hmd` array)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Lines 3+: one field per record.

#	Column	Type	Description
1	filename	char(50)	Name of a `*.hmd` time-series file. Used as the station key (`hmd_n(i)`).

Example

```
hmd.cli
FILENAME
ames.hmd
```

wnd.cli

List of measured wind speed station files.

Source

- Reader: src/cli_wmeas.f90
- Type: src/climate_module.f90, `type climate_measured_data` (`wnd` array)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Lines 3+: one field per record.

#	Column	Type	Description
1	filename	char(50)	Name of a `*.wnd` time-series file. Used as the station key (`wnd_n(i)`).

Example

```
wnd.cli
FILENAME
ames.wnd
```

pet.cli

List of measured potential ET station files. Used when codes.bsn sets `pet = 3`.

Source

- Reader: src/cli_petmeas.f90
- Type: src/climate_module.f90, `type climate_measured_data` (`petm` array)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Lines 3+: one field per record.

#	Column	Type	Description
1	filename	char(50)	Name of a `*.pet` time-series file. Used as the station key (`petm_n(i)`).

Example

```
pet.cli
FILENAME
ames.pet
```

atmodep.cli

Atmospheric N deposition for one or more stations. Unlike the per-variable index files, this file contains the deposition values themselves, not pointers to data files.

Source

- Reader: src/cli_read_atmodep.f90
- Types: src/climate_module.f90, `type atmospheric_deposition_control` (line 3) and `type atmospheric_deposition` (per-station block)

Format

- Line 1: title (skipped).
- Line 2: header (skipped).
- Line 3: control record, 5 fields.
- Then, per station, a block whose layout depends on `timestep`.

Control record (line 3):

#	Field	Type	Description
1	num_sta	int	Number of deposition stations to read.
2	timestep	char(2)	`aa` (average annual, scalar values), `mo` (monthly time series), or `yr` (annual time series).
3	mo_init	int	First month of the time series (1 to 12). Used when `timestep = mo`.
4	yr_init	int	First year of the time series. Used for `mo` and `yr`.
5	num	int	Number of records per station (length of the time series). Set to `1` for `aa`.

For `timestep = aa`, each station occupies 5 lines:

Line	Type	Description
1	char(50)	Station name (used as `atmo_n(i)`).
2	real	`nh4_rf` - average annual ammonia concentration in rainfall (mg/L).
3	real	`no3_rf` - average annual nitrate concentration in rainfall (mg/L).
4	real	`nh4_dry` - average annual ammonia dry deposition (kg/ha/yr).
5	real	`no3_dry` - average annual nitrate dry deposition (kg/ha/yr).

For `timestep = mo`, each station occupies 5 lines (name plus four monthly series of length `num`):

Line	Description
1	Station name.
2	`nh4_rfmo(1..num)` - ammonia in rainfall by month (mg/L).
3	`no3_rfmo(1..num)` - nitrate in rainfall by month (mg/L).
4	`nh4_drymo(1..num)` - ammonia dry deposition by month (kg/ha/yr).
5	`no3_drymo(1..num)` - nitrate dry deposition by month (kg/ha/yr).

For `timestep = yr`, each station occupies 5 lines (name plus four annual series of length `num`):

Line	Description
1	Station name.
2	`nh4_rfyr(1..num)` - ammonia in rainfall by year (mg/L).
3	`no3_rfyr(1..num)` - nitrate in rainfall by year (mg/L).
4	`nh4_dryyr(1..num)` - ammonia dry deposition by year (kg/ha/yr).
5	`no3_dryyr(1..num)` - nitrate dry deposition by year (kg/ha/yr).

When `num_sta = 0` (no stations) or the file is named `null` in `file.cio` or absent, the reader allocates an empty array and skips deposition.

Example

Minimal `aa` form, one station:

```
atmodep.cli
num_sta  timestep  mo_init  yr_init  num
      1        aa        1     2000    1
station_001
1.0
0.2
0.0
0.0
```

Atmospheric deposition

Three readers populate the atmospheric deposition state. They share the per-station list set up by `atmodep.cli` and the control record `atmodep_cont` (`type atmospheric_deposition_control`). The three files cover N, salts, and other constituents.

atmodep.cli (nitrogen)

Documented above. Read by src/cli_read_atmodep.f90. Fills `atmodep(0:num_sta)` (`type atmospheric_deposition`) with NH4 and NO3 wet (rainfall) concentrations and dry deposition rates. The reader also computes `atmodep_cont%ts` (the index into the time series corresponding to the first simulation step) for `mo` and `yr` time steps, and sets `atmodep_cont%first = 0` once aligned.

salt_atmo.cli (salt ions)

Read by src/cli_read_atmodep_salt.f90 when `cs_db%num_salts > 0`. Per-station wet and dry deposition for each salt ion in `constituents.cs`. Filled into `atmodep_salt(iadep)%salt(isalt)` (`type atmospheric_deposition_cs` within `type object_deposition_cs`).

Layout:

- Lines 1 to 6: skipped. The reader consumes six lines as commentary before processing stations.
- Per station, block layout depends on `atmodep_cont%timestep` (set by `atmodep.cli`):

For `timestep = aa`:

Line	Description
1	Station name (character, ignored for cross-matching; positional).
2..1+num_salts	`salt_ion_label`, then one real: `rf` (rainfall concentration, mg/L) for each ion.
…next num_salts	`salt_ion_label`, then one real: `dry` (dry deposition, kg/ha/yr) for each ion.

For `timestep = mo`:

Line	Description
1	Station name.
2..1+num_salts	`salt_ion_label`, then `num` reals: monthly wet concentrations (mg/L).
…next num_salts	`salt_ion_label`, then `num` reals: monthly dry deposition (kg/ha/yr).

For `timestep = yr`: same shape as monthly with annual series of length `num`.

When the file exists, the reader sets the module-level flag `salt_atmo = "y"`. When `cs_db%num_salts = 0` or the file is missing, salt deposition is skipped.

cs_atmo.cli (other constituents)

Read by src/cli_read_atmodep_cs.f90 when `cs_db%num_cs > 0`. Same shape as `salt_atmo.cli` but without per-ion labels.

Layout:

- Lines 1 to 3: skipped.
- Per station:

For `timestep = aa`:

Line	Description
1	Station name (real-typed scratch read; positional).
2..1+num_cs	One real per line: wet concentration (mg/L) for each constituent.
...next num_cs	One real per line: dry deposition (kg/ha) for each constituent.

For `timestep = mo` and `timestep = yr`, each constituent gets one line of `num` reals for wet, and one line of `num` reals for dry, in the same order as `constituents.cs`.

When the file exists, the reader sets `cs_atmo = "y"`.

Related files

- file.cio. Names every `.cli` file via `type input_cli`.
- codes.bsn. `pet = 3` triggers reading `pet.cli`.
- `*.pcp`, `*.tmp`, `*.slr`, `*.hmd`, `*.wnd`, `*.pet`. The actual time-series files listed in the per-variable `.cli` files.
- salt module, cs module. Define the salt and other-constituent lists that drive `salt_atmo.cli` and `cs_atmo.cli`.

Purpose

A `*.pcp` file holds the measured precipitation time series for a single gauge. One file per station. The filename is listed in pcp.cli and the same filename string is matched as the gauge key referenced from `pgage` in weather-sta.cli. The reader supports either daily values or subdaily values at a fixed time step.

Source

- Reader: src/cli_pmeas.f90
- Type definition: src/climate_module.f90, `type climate_measured_data` (the `pcp` array)

Format

The reader uses list-directed reads (free format, whitespace-separated). The first two lines are skipped. The third line is the station header. From line four to end of file are the time-step records.

Line 1: title

Single string, skipped.

Line 2: column header

Skipped. SWAT+ Editor writes `NBYR TSTEP LAT LONG ELEV`.

Line 3: station header

Read by `read (108,*,iostat=eof) pcp(i)%nbyr, pcp(i)%tstep, pcp(i)%lat, pcp(i)%long, pcp(i)%elev`.

#	Field	Type	Units	Description
1	nbyr	int	years	Number of years of record in the file. Used to size the in-memory time-series array.
2	tstep	int	minutes	Time step. `0` means daily. A positive value is the subdaily interval in minutes and must match `time%step` from `time.sim`.
3	lat	real	degrees	Latitude of the gauge.
4	long	real	degrees	Longitude of the gauge.
5	elev	real	m	Elevation of the gauge.

Records (line 4 onward)

The record layout depends on `tstep`.

For daily records (`tstep = 0`):

#	Field	Type	Units	Description
1	year	int	calendar year	Calendar year of the record.
2	jday	int	day-of-year	Julian day (1 to 365 or 366).
3	pcp	real	mm	Daily precipitation depth. Use a value at or below `-97` to flag missing data; the reader replaces it with a generated value.

For subdaily records (`tstep > 0`):

#	Field	Type	Units	Description
1	year	int	calendar year	Calendar year.
2	jday	int	day-of-year	Julian day.
3	mo	int	month	Month number.
4	day_mo	int	day	Day of the month.
5	ihr	int	step index	Sub-day step index (1 to `1440 / tstep`).
6	pcp	real	mm	Precipitation depth for the time step.

Records are read in order. The reader advances year-by-year using the change in `jday` (it expects the year to change after `jday = 365` or `366`).

Missing data

The daily reader treats any value at or below `-97` as missing. Downstream in `cli_precip_control.f90`, when `precip_next <= -97.` the value is replaced by the weather generator. The classic flag written by SWAT+ Editor and the legacy tools is `-99`.

Example

First lines of `refdata/Ames_sub1/ames.pcp`:

```
AME.pcp
 NBYR    TSTEP      LAT      LONG     ELEV
   121        0    42.04    -93.89   316
1900 1 -99
1900 2 -99
1900 3 -99
```

```
...
1983 184 0
1983 185 15.5
1983 186 0
```

Station header: 121 years, daily time step (`0`), latitude 42.04, longitude -93.89, elevation 316 m. Early years are missing (`-99`) and will be filled by the weather generator.

Related files

- `pcp.cli`. Lists each `*.pcp` filename.
- `weather-sta.cli`. Binds a precipitation gauge to a weather station via `pgage`.
- `file.cio`. Names the `pcp.cli` index and the optional path prefix `in_path_pcp%pcp` applied to every `*.pcp` filename.
- `time.sim`. For subdaily files, `time%step` must equal the file's `tstep`.

Purpose

A `*.tmp` file holds the measured daily temperature time series for a single gauge. One file per station. The filename is listed in `tmp.cli` and the same filename string is matched as the gauge key referenced from `tgage` in `weather-sta.cli`. The reader records daily maximum and minimum air temperature.

Source

- Reader: `src/cli_tmeas.f90`
- Type definition: `src/climate_module.f90`, `type climate_measured_data` (the `tmp` array; `ts` holds daily max, `ts2` holds daily min)

Format

List-directed reads (free format, whitespace-separated). The first two lines are skipped. The third line is the station header. From line four to end of file are the daily records.

Line 1: title

Single string, skipped.

Line 2: column header

Skipped. SWAT+ Editor writes `NBYR  TSTEP  LAT  LONG  ELEV`.

Line 3: station header

Read by `read (108,*,iostat=eof) tmp(i)%nbyr, tmp(i)%tstep, tmp(i)%lat, tmp(i)%long, tmp(i)%elev`.

#	Field	Type	Units	Description
1	nbyr	int	years	Number of years of record in the file.
2	tstep	int	minutes	Time step indicator. Stored but the reader expects daily values regardless.
3	lat	real	degrees	Latitude of the gauge.
4	long	real	degrees	Longitude of the gauge.
5	elev	real	m	Elevation of the gauge.

Records (line 4 onward)

Daily values only:

#	Field	Type	Units	Description
1	year	int	calendar year	Calendar year of the record.
2	jday	int	day-of-year	Julian day (1 to 365 or 366).
3	tmax	real	deg C	Daily maximum air temperature. Stored in `tmp(i)%ts(jday, year_index)`.
4	tmin	real	deg C	Daily minimum air temperature. Stored in `tmp(i)%ts2(jday, year_index)`.

Records are read sequentially. The reader detects the year change after `jday = 365` or `366` and increments its internal year index.

While reading, the routine accumulates monthly sums into `tmp(i)%max_mon` and `tmp(i)%min_mon` (per the `xmon` call) and at end of file divides by the total day count to get the average monthly maximum and minimum. A `-99` flag is treated like any other value at this stage, so missing-data flags carry into the monthly statistics unless the file is clean over the simulation window.

Example

First lines of `refdata/Ames_sub1/ames.tem` (the temperature time series for the Ames reference dataset; the same record layout applies to a `*.tmp` file):

```
AME.Tmp
 NBYR    TSTEP       LAT      LONG      ELEV
   121        0     42.04   -93.89    316
1900    1   -99 -99
1900    2   -99 -99
...
1983    185 32.2    10.6
1983    186 26.7    11.1
```

Station header: 121 years, time step `0`, latitude 42.04, longitude -93.89, elevation 316 m. Early years carry the `-99` missing-data flag for both max and min.

Related files

- `tmp.cli`. Lists each temperature gauge filename.
- `weather-sta.cli`. Binds a temperature gauge to a weather station via `tgage`.
- `file.cio`. Names the `tmp.cli` index and the optional path prefix `in_path_tmp%tmp` applied to every temperature filename.
- `*.tem`. Same record layout, alternate filename extension used by some toolchains.

Purpose

A `*.tem` file is a temperature time-series file with the same record layout as a `*.tmp`. The `.tem` extension is used by the reference datasets shipped with SWAT+ (for example `refdata/Ames_sub1/ames.tem`) and by some external preprocessors. There is no separate reader for `.tem`. The temperature reader at `src/cli_tmeas.f90` opens whatever filename is supplied; the extension is not inspected.

Source

- Reader: `src/cli_tmeas.f90`
- Type definition: `src/climate_module.f90`, `type climate_measured_data`

The reader is reached only when the filename appears in tmp.cli. The tgage slot in weather-sta.cli is matched against the filename listed in tmp.cli via the tmp_n lookup array, so both files must agree on the string (including the .tem extension if used).

Format

Identical to a daily *.tmp file. List-directed reads, whitespace-separated.

- Line 1: title (skipped).
- Line 2: column header (skipped). Conventionally NBYR TSTEP LAT LONG ELEV.
- Line 3: station header (nbyr, tstep, lat, long, elev).
- Line 4 onward: one record per day with year, Julian day, daily maximum temperature, daily minimum temperature.

See *.tmp for the field tables.

Example

refdata/Ames_sub1/ames.tem (excerpt):

```
AME.Tmp
 NBYR    TSTEP      LAT      LONG     ELEV
   121        0    42.04   -93.89   316
1900    1   -99 -99
1900    2   -99 -99
...
1983    185 32.2    10.6
1983    186 26.7    11.1
```

Related files

- *.tmp. Same content, alternate extension. The reader is shared.
- tmp.cli. Lists each temperature filename (with whatever extension).
- weather-sta.cli. The tgage column must match a filename entry in tmp.cli.

Purpose

snow.sno is the snow parameter database. Each row defines a named set of snow accumulation and melt parameters. HRUs reference a row by name through the snow column of hru-data.hru, which the reader resolves to an index into the snodb array. If hru-data.hru carries null for an HRU's snow entry, that HRU uses the default snow_parameters values.

Source

- Reader: src/snowdb_read.f90
- Type definition: src/hru_module.f90, type snow_parameters (the snodb array)
- Filename slot: in_parmdb%snow in src/input_file_module.f90 (default snow.sno).

Format

List-directed reads (free format, whitespace-separated). The first two lines are skipped. From line three to end of file each line is one snow parameter row. The reader counts records on a first pass, allocates snodb(0:imax), then reads each row into snodb(isno).

- Line 1: title (skipped). Editor convention: snow.sno: written by SWAT+ editor ....

- Line 2: column header (skipped). Editor convention: `name fall_tmp melt_tmp melt_max melt_min tmp_lag snow_h2o cov50 snow_init`.
- Line 3+: one row per parameter set, 9 fields each.

The fields appear in the order declared in `type snow_parameters`:

#	Field	Type	Units	Default	Description
1	name	char(40)	-	(none)	Parameter set name. Referenced from the `snow` column of `hru-data.hru`.
2	falltmp	real	deg C	0.0	Snowfall temperature. Precipitation falls as snow when the mean air temperature is at or below this threshold.
3	melttmp	real	deg C	0.5	Snowmelt base temperature. Snowmelt only occurs when the snowpack temperature exceeds this value.
4	meltmx	real	mm/deg C/day	4.5	Maximum melt rate factor for snow during the year (occurs near 21 June).
5	meltmn	real	mm/deg C/day	0.5	Minimum melt rate factor for snow during the year (occurs near 21 December).
6	timp	real	none	0.8	Snowpack temperature lag factor (0 to 1). Higher values give more weight to the current day's air temperature.
7	covmx	real	mm H2O	25.0	Snow water content above which the ground is fully covered by snow.
8	cov50	real	none	0.5	Fraction of `covmx` at which the ground is 50% snow-covered. Shape factor for the snowpack areal depletion curve.
9	init_mm	real	mm H2O	0.0	Initial snow water content at the start of the simulation.

The defaults above are the values assigned in `type snow_parameters` at declaration. They are not values the reader supplies, so every row of `snow.sno` must list all nine fields.

Example

`refdata/Ames_sub1/snow.sno`:

```
snow.sno: written by SWAT+ editor v2.3.3 on 2023-10-25 08:58 for SWAT+ rev.60.5.7
name                  fall_tmp      melt_tmp      melt_max      melt_min       tmp_lag      snow_h2o
snow001                1.00000       0.50000       4.50000       4.50000       1.00000       1.00000
```

One parameter set named `snow001`. Note that the example value for `meltmn` (column 5) is `4.5`, identical to `meltmx`. The defaults shipped in source set `meltmn = 0.5`.

Related files

- hru-data.hru. References a `snow.sno` row by name in its `snow` column. `null` selects the type defaults.
- file.cio. Names the `snow.sno` file in the `init` group via `in_parmdb%snow`. Set to `null` to skip reading (no snow database is loaded; HRUs that point at a snow name will fall back as described above).